

Localo Complete Database Blueprint

End-to-end backend database design for Localo: users, admin, shop owners, shops, products, search, commissions, support, notifications, orders-ready foundations, audit, and operations.

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Recommended stack assumption: PostgreSQL + Prisma/NestJS or Node backend. The schema uses UUID primary keys, timestampz, JSONB for flexible metadata, soft deletes for business records, and normalized relational tables for reporting and long-term scale.

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1. Database Strategy

- **PostgreSQL first:** best fit for relational joins, JSONB, full-text search, indexing, and analytics later.
- **UUID primary keys:** every main table uses id UUID PK for safer distributed creation and public APIs.
- **Audit columns:** created_at, updated_at, deleted_at, created_by_id, updated_by_id where useful.
- **Soft delete:** business-critical tables use deleted_at instead of hard delete.
- **Multi-role user model:** one user can be admin, shop owner, staff, and customer without duplicate accounts.
- **Backend first:** build DB, APIs, auth, seed data, and admin operations before mobile/shop UI polish.

Global Naming and Column Standards

Pattern	Use
id UUID PK	Primary key on all main tables.
*_id UUID FK	Foreign key columns use suffix _id.
status VARCHAR/ENUM	Use app-level enum values documented in this PDF.
metadata JSONB	For low-risk extension data that should not become dozens of columns too early.
created_at / updated_at timestamptz	Always store UTC timestamps.
deleted_at timestamptz nullable	Soft delete. Create partial unique indexes using WHERE deleted_at IS NULL.
money fields numeric(12,2)	Avoid float for prices, commission, tax, invoices.
latitude / longitude numeric(10,7)	Good enough precision for store location search.

2. High-Level Entity Relationship Map

Core relationship flow:

users -> shop_team_members -> shops -> products -> variants/prices/inventory
shops -> commission_settings -> commission_invoices -> invoice_items/payments
users -> favorites/reviews/orders/support_tickets/notifications
categories -> shop_categories and product_category_links
shops -> locations/business_hours/media/documents/status_history

Domain	Main tables	Why it exists
Identity & RBAC	users, user_sessions, roles, permissions, user_roles	One account can hold multiple responsibilities.
Location	countries, states, cities, areas, user_addresses, shop_locations	Location search, delivery-ready future, clean address UX.
Shop Management	shops, shop_profiles, shop_team_members, business_hours, special_hours, documents	Admin creates/approves shops; owners manage shop data.
Catalog	categories, brands, products, variants, prices, inventory, media, attributes	Searchable product catalog with long-term flexibility.
Discovery	search_queries, favorites, recently_viewed_items, shop_views	User engagement and recommendations later.

Domain	Main tables	Why it exists
Orders-ready	carts, cart_items, orders, order_items, payment_transactions	Not required for day one, but backend shape supports growth.
Commission & Billing	commission_plans, commission_settings, invoices, invoice_items, payments	Platform monetization from shops.
Support	support_tickets, support_messages, support_attachments	Shop owner/user support workflow.
Reviews	reviews, review_replies, review_reports	Trust layer for shops/products.
System & Audit	audit_logs, app_settings, feature_flags, webhook_events	Operations, debugging, compliance.

3. Complete Table Catalogue

Identity & Access

users

Main user account for admin, shop owner, staff, and customer.

Column	Type	Constraint	Purpose
id	uuid	PK	Primary user id
full_name	varchar(160)	not null	Display/legal name
email	varchar(255)	unique nullable	Login/contact email
phone	varchar(30)	unique nullable	Login/contact phone
password_hash	text	nullable	Null if OAuth only
avatar_media_id	uuid	FK media_assets nullable	Profile photo
user_type	varchar(30)	default customer	customer/admin/shop_owner/staff/support
status	varchar(30)	default active	active/pending/suspended/deleted
email_verified_at	timestampz	nullable	Email verified timestamp
phone_verified_at	timestampz	nullable	Phone verified timestamp
last_login_at	timestampz	nullable	Last successful login
metadata	jsonb	default {}	Preferences/extra profile data
created_at	timestampz	not null	Created time
updated_at	timestampz	not null	Updated time
deleted_at	timestampz	nullable	Soft delete

Indexes / Constraints
unique lower(email) where email is not null and deleted_at is null
unique phone where phone is not null and deleted_at is null
index status
index user_type

user_sessions

Refresh/session tokens and login devices.

Column	Type	Constraint	Purpose
id	uuid	PK	Session id
user_id	uuid	FK users not null	Owner user
refresh_token_hash	text	not null	Hash only, never raw token
device_name	varchar(120)	nullable	Browser/mobile name
ip_address	inet	nullable	Last IP
user_agent	text	nullable	User agent
expires_at	timestampz	not null	Session expiry
revoked_at	timestampz	nullable	Logout/revoke
created_at	timestampz	not null	Created time
Indexes / Constraints			
index user_id			
index expires_at			
index revoked_at			

roles

Role catalog.

Column	Type	Constraint	Purpose
id	uuid	PK	Role id
code	varchar(60)	unique	admin, shop_owner, staff, customer
name	varchar(120)	not null	Readable role name
description	text	nullable	Role purpose
is_system	boolean	default false	Cannot delete system roles
created_at	timestampz	not null	Created time
Indexes / Constraints			
unique code			

permissions

Permission catalog used by backend guards.

Column	Type	Constraint	Purpose
id	uuid	PK	Permission id

Column	Type	Constraint	Purpose
code	varchar(120)	unique	Example: shops.approve
module	varchar(80)	not null	shops/products/support
description	text	nullable	Meaning
created_at	timestampz	not null	Created time

Indexes / Constraints
unique code
index module

user_roles

Many-to-many users to roles.

Column	Type	Constraint	Purpose
id	uuid	PK	Mapping id
user_id	uuid	FK users	User
role_id	uuid	FK roles	Role
shop_id	uuid	FK shops nullable	Scoped role for a shop
assigned_by_id	uuid	FK users nullable	Admin/owner who assigned
created_at	timestampz	not null	Created time

Indexes / Constraints
unique user_id, role_id, shop_id
index user_id
index shop_id

role_permissions

Many-to-many roles to permissions.

Column	Type	Constraint	Purpose
id	uuid	PK	Mapping id
role_id	uuid	FK roles	Role
permission_id	uuid	FK permissions	Permission
created_at	timestampz	not null	Created time

Indexes / Constraints
unique role_id, permission_id

media_assets

Reusable uploaded files/images.

Column	Type	Constraint	Purpose
id	uuid	PK	Media id
owner_user_id	uuid	FK users nullable	Uploader
storage_provider	varchar(40)	not null	s3/r2/local
bucket	varchar(120)	nullable	Storage bucket
path	text	not null	Object key/path
public_url	text	nullable	CDN URL
mime_type	varchar(120)	not null	image/jpeg etc
size_bytes	bigint	nullable	File size
width	int	nullable	Image width
height	int	nullable	Image height
alt_text	varchar(255)	nullable	Accessibility text
metadata	jsonb	default {}	Extra file info
created_at	timestampz	not null	Created time
Indexes / Constraints			
index owner_user_id			
index mime_type			

Location & Address

countries

Country reference table.

Column	Type	Constraint	Purpose
id	uuid	PK	Country id
iso2	char(2)	unique	IN
name	varchar(120)	not null	Country name
currency_code	char(3)	not null	INR/USD
phone_code	varchar(8)	nullable	+91
is_active	boolean	default true	Enabled country
Indexes / Constraints			
unique iso2			

states

State/province reference.

Column	Type	Constraint	Purpose
id	uuid	PK	State id
country_id	uuid	FK countries	Country
name	varchar(120)	not null	State name
code	varchar(20)	nullable	KA
is_active	boolean	default true	Enabled
Indexes / Constraints			
index country_id			
unique country_id, name			

cities

City reference.

Column	Type	Constraint	Purpose
id	uuid	PK	City id
state_id	uuid	FK states	State
name	varchar(120)	not null	City name
slug	varchar(140)	not null	SEO/search slug
is_active	boolean	default true	Enabled
Indexes / Constraints			
index state_id			
unique state_id, slug			

areas

Area/locality reference for better search.

Column	Type	Constraint	Purpose
id	uuid	PK	Area id
city_id	uuid	FK cities	City
name	varchar(140)	not null	Area/locality name
slug	varchar(160)	not null	Search slug
pincode	varchar(20)	nullable	Postal code
latitude	numeric(10,7)	nullable	Area center lat

Column	Type	Constraint	Purpose
longitude	numeric(10,7)	nullable	Area center lng
is_active	boolean	default true	Enabled

Indexes / Constraints			
index city_id			
index pincode			
unique city_id, slug			

user_addresses

Saved addresses for customers/shop owners.

Column	Type	Constraint	Purpose
id	uuid	PK	Address id
user_id	uuid	FK users	Owner
label	varchar(80)	nullable	Home/work
line1	varchar(255)	not null	Street line 1
line2	varchar(255)	nullable	Street line 2
area_id	uuid	FK areas nullable	Area
city_id	uuid	FK cities nullable	City
state_id	uuid	FK states nullable	State
country_id	uuid	FK countries nullable	Country
pincode	varchar(20)	nullable	Postal code
latitude	numeric(10,7)	nullable	Lat
longitude	numeric(10,7)	nullable	Lng
is_default	boolean	default false	Default address
created_at	timestampz	not null	Created
updated_at	timestampz	not null	Updated
deleted_at	timestampz	nullable	Soft delete

Indexes / Constraints			
index user_id			
index area_id			
index latitude, longitude			

Shop Management

shops

Primary shop/business entity.

Column	Type	Constraint	Purpose
id	uuid	PK	Shop id
owner_user_id	uuid	FK users not null	Primary owner
name	varchar(180)	not null	Shop display name
slug	varchar(220)	unique	SEO/public slug
legal_name	varchar(220)	nullable	Registered name
description	text	nullable	Shop description
primary_category_id	uuid	FK categories nullable	Main category
status	varchar(40)	default draft	draft/pending_review/active/paused/suspended/rejected/closed
verification_status	varchar(40)	default unverified	unverified/pending/verified/rejected
commission_status	varchar(40)	default not_configured	not_configured/active/overdue/paused
logo_media_id	uuid	FK media_assets nullable	Shop logo
cover_media_id	uuid	FK media_assets nullable	Cover image
phone	varchar(30)	nullable	Primary phone
email	varchar(255)	nullable	Shop email
website_url	text	nullable	Website
rating_avg	numeric(3,2)	default 0	Cached average rating
rating_count	int	default 0	Cached rating count
is_featured	boolean	default false	Featured shop
created_by_admin_id	uuid	FK users nullable	Admin who created shop
approved_by_id	uuid	FK users nullable	Approver
approved_at	timestampz	nullable	Approval time
metadata	jsonb	default {}	Extra settings
created_at	timestampz	not null	Created
updated_at	timestampz	not null	Updated
deleted_at	timestampz	nullable	Soft delete

Indexes / Constraints
unique slug where deleted_at is null
index owner_user_id
index status
index primary_category_id
index is_featured
index rating_avg

shop_locations

Physical shop locations.

Column	Type	Constraint	Purpose
id	uuid	PK	Location id
shop_id	uuid	FK shops not null	Shop
label	varchar(100)	default Main	Branch name
address_line1	varchar(255)	not null	Address line 1
address_line2	varchar(255)	nullable	Address line 2
area_id	uuid	FK areas nullable	Area
city_id	uuid	FK cities nullable	City
state_id	uuid	FK states nullable	State
country_id	uuid	FK countries nullable	Country
pincode	varchar(20)	nullable	Postal code
latitude	numeric(10,7)	not null	Lat for map
longitude	numeric(10,7)	not null	Lng for map
google_place_id	varchar(255)	nullable	Google place id
is_primary	boolean	default true	Primary location
is_service_area_enabled	boolean	default false	Future delivery/service radius
service_radius_km	numeric(6,2)	nullable	Radius
created_at	timestamptz	not null	Created
updated_at	timestamptz	not null	Updated
Indexes / Constraints			
index shop_id			
index area_id			
GIST index geography(point) for radius search			
partial unique shop_id where is_primary=true			

shop_business_hours

Weekly business hours.

Column	Type	Constraint	Purpose
id	uuid	PK	Hours id
shop_id	uuid	FK shops	Shop
day_of_week	smallint	0-6	0 Sunday, 6 Saturday
opens_at	time	nullable	Opening time
closes_at	time	nullable	Closing time

Column	Type	Constraint	Purpose
is_closed	boolean	default false	Closed all day
breaks	jsonb	default []	Optional break slots
created_at	timestampz	not null	Created
updated_at	timestampz	not null	Updated
Indexes / Constraints			
unique shop_id, day_of_week			
index shop_id			

shop_special_hours

Holiday/special day overrides.

Column	Type	Constraint	Purpose
id	uuid	PK	Special hours id
shop_id	uuid	FK shops	Shop
date	date	not null	Special date
opens_at	time	nullable	Override open
closes_at	time	nullable	Override close
is_closed	boolean	default false	Closed on date
reason	varchar(160)	nullable	Festival/holiday reason
created_at	timestampz	not null	Created
Indexes / Constraints			
unique shop_id, date			

shop_team_members

Shop staff and owner access.

Column	Type	Constraint	Purpose
id	uuid	PK	Team id
shop_id	uuid	FK shops	Shop
user_id	uuid	FK users	Member
role	varchar(40)	not null	owner/manager/staff/viewer
status	varchar(30)	default active	invited/active/removed
invited_by_id	uuid	FK users nullable	Inviter
joined_at	timestampz	nullable	Join time
created_at	timestampz	not null	Created

Column	Type	Constraint	Purpose
deleted_at	timestampz	nullable	Soft delete
Indexes / Constraints			
unique shop_id, user_id where deleted_at is null			
index user_id			
index role			

shop_documents

Verification/KYC documents.

Column	Type	Constraint	Purpose
id	uuid	PK	Document id
shop_id	uuid	FK shops	Shop
document_type	varchar(60)	not null	gst/license/pan/address_proof/other
document_number	varchar(120)	nullable	Document number
media_id	uuid	FK media_assets	Uploaded file
status	varchar(40)	default pending	pending/approved/rejected/expired
reviewed_by_id	uuid	FK users nullable	Admin reviewer
reviewed_at	timestampz	nullable	Review time
rejection_reason	text	nullable	Reason
expires_at	date	nullable	Expiry
created_at	timestampz	not null	Created
Indexes / Constraints			
index shop_id			
index document_type			
index status			

shop_status_history

Every status change for admin audit.

Column	Type	Constraint	Purpose
id	uuid	PK	History id
shop_id	uuid	FK shops	Shop
from_status	varchar(40)	nullable	Old status
to_status	varchar(40)	not null	New status
reason	text	nullable	Why changed

Column	Type	Constraint	Purpose
changed_by_id	uuid	FK users nullable	Actor
created_at	timestampz	not null	Changed time
Indexes / Constraints			
index shop_id			
index created_at			

Category & Catalog

categories

Hierarchical category tree for shops and products.

Column	Type	Constraint	Purpose
id	uuid	PK	Category id
parent_id	uuid	FK categories nullable	Parent category
name	varchar(160)	not null	Category name
slug	varchar(180)	not null	Slug
description	text	nullable	Description
icon_media_id	uuid	FK media_assets nullable	Icon
image_media_id	uuid	FK media_assets nullable	Image
level	smallint	default 0	Depth
sort_order	int	default 0	Ordering
is_active	boolean	default true	Enabled
metadata	jsonb	default {}	Filters/extra
created_at	timestampz	not null	Created
updated_at	timestampz	not null	Updated
Indexes / Constraints			
unique parent_id, slug			
index parent_id			
index is_active			
index sort_order			

shop_categories

Many-to-many shop category mapping.

Column	Type	Constraint	Purpose
id	uuid	PK	Mapping id
shop_id	uuid	FK shops	Shop
category_id	uuid	FK categories	Category
is_primary	boolean	default false	Primary shop category
created_at	timestampz	not null	Created

Indexes / Constraints
unique shop_id, category_id
index category_id
partial unique shop_id where is_primary=true

product_brands

Optional product brand catalog.

Column	Type	Constraint	Purpose
id	uuid	PK	Brand id
name	varchar(160)	not null	Brand name
slug	varchar(180)	unique	Brand slug
logo_media_id	uuid	FK media_assets nullable	Logo
is_active	boolean	default true	Enabled
created_at	timestampz	not null	Created

Indexes / Constraints
unique slug
index name

products

Product/service listed by a shop.

Column	Type	Constraint	Purpose
id	uuid	PK	Product id
shop_id	uuid	FK shops	Owner shop
brand_id	uuid	FK product_brands nullable	Brand
title	varchar(220)	not null	Product title
slug	varchar(260)	not null	Shop-scoped slug
description	text	nullable	Long description
short_description	varchar(500)	nullable	Card/list text

Column	Type	Constraint	Purpose
product_type	varchar(40)	default product	product/service/digital
status	varchar(40)	default draft	draft/active/out_of_stock/hidden/archived/rejected
sku	varchar(100)	nullable	Shop SKU
search_keywords	text[]	default {}	Extra search terms
tags	text[]	default {}	Fast simple tags
base_price	numeric(12,2)	nullable	Cached starting price
compare_at_price	numeric(12,2)	nullable	MRP/strike price
currency_code	char(3)	default INR	Currency
tax_rate	numeric(5,2)	default 0	GST/tax percent
is_featured	boolean	default false	Featured product
rating_avg	numeric(3,2)	default 0	Cached rating
rating_count	int	default 0	Cached reviews
published_at	timestamptz	nullable	Live time
metadata	jsonb	default {}	Flexible specs
created_at	timestamptz	not null	Created
updated_at	timestamptz	not null	Updated
deleted_at	timestamptz	nullable	Soft delete
Indexes / Constraints			
unique shop_id, slug where deleted_at is null			
index shop_id, status			
index brand_id			
GIN index tags			
GIN full-text index title/description/search_keywords			
index is_featured			
index base_price			

product_category_links

Many-to-many product category mapping.

Column	Type	Constraint	Purpose
id	uuid	PK	Mapping id
product_id	uuid	FK products	Product
category_id	uuid	FK categories	Category
is_primary	boolean	default false	Primary category
created_at	timestamptz	not null	Created

Indexes / Constraints
unique product_id, category_id
index category_id
partial unique product_id where is_primary=true

product_variants

Variants such as size/color/weight.

Column	Type	Constraint	Purpose
id	uuid	PK	Variant id
product_id	uuid	FK products	Product
title	varchar(180)	not null	Variant name
sku	varchar(120)	nullable	Variant SKU
barcode	varchar(120)	nullable	Barcode
options	jsonb	default {}	{size,color,weight}
price	numeric(12,2)	not null	Selling price
compare_at_price	numeric(12,2)	nullable	MRP
cost_price	numeric(12,2)	nullable	Internal cost
is_default	boolean	default false	Default variant
status	varchar(40)	default active	active/hidden/out_of_stock
created_at	timestamptz	not null	Created
updated_at	timestamptz	not null	Updated
deleted_at	timestamptz	nullable	Soft delete

Indexes / Constraints
index product_id
unique product_id, sku where sku is not null and deleted_at is null
partial unique product_id where is_default=true and deleted_at is null

product_inventory

Inventory per variant and optional location.

Column	Type	Constraint	Purpose
id	uuid	PK	Inventory id
product_id	uuid	FK products	Product
variant_id	uuid	FK product_variants nullable	Variant
shop_location_id	uuid	FK shop_locations nullable	Branch

Column	Type	Constraint	Purpose
quantity_available	int	default 0	Available stock
quantity_reserved	int	default 0	Reserved stock
low_stock_threshold	int	default 5	Warning threshold
track_inventory	boolean	default true	Track stock
updated_at	timestampz	not null	Updated

Indexes / Constraints
unique variant_id, shop_location_id
index product_id
index quantity_available

product_media

Product images/videos.

Column	Type	Constraint	Purpose
id	uuid	PK	Media link id
product_id	uuid	FK products	Product
variant_id	uuid	FK product_variants nullable	Variant-specific
media_id	uuid	FK media_assets	Media
sort_order	int	default 0	Ordering
is_primary	boolean	default false	Main image
created_at	timestampz	not null	Created

Indexes / Constraints
index product_id
index variant_id
partial unique product_id where is_primary=true

product_attribute_definitions

Category/product filter definitions.

Column	Type	Constraint	Purpose
id	uuid	PK	Attribute id
category_id	uuid	FK categories nullable	Optional category scope
code	varchar(80)	not null	material/size/color
label	varchar(120)	not null	Readable label
data_type	varchar(30)	not null	text/number/boolean/select/multi_select

Column	Type	Constraint	Purpose
options	jsonb	default []	Allowed options
is_filterable	boolean	default false	Show in filters
is_required	boolean	default false	Required
sort_order	int	default 0	Ordering
created_at	timestampz	not null	Created
Indexes / Constraints			
unique category_id, code			
index is_filterable			

product_attribute_values

Actual product attribute values.

Column	Type	Constraint	Purpose
id	uuid	PK	Value id
product_id	uuid	FK products	Product
attribute_id	uuid	FK attribute_definitions	Attribute
value_text	text	nullable	Text value
value_number	numeric(12,2)	nullable	Number value
value_boolean	boolean	nullable	Boolean value
value_json	jsonb	nullable	Complex value
created_at	timestampz	not null	Created
Indexes / Constraints			
unique product_id, attribute_id			
index attribute_id			
index value_text			

Discovery, Reviews & User Engagement

user_favorites

Saved shops/products by users.

Column	Type	Constraint	Purpose
id	uuid	PK	Favorite id
user_id	uuid	FK users	User
shop_id	uuid	FK shops nullable	Favorite shop

Column	Type	Constraint	Purpose
product_id	uuid	FK products nullable	Favorite product
created_at	timestampz	not null	Created

Indexes / Constraints			
unique user_id, shop_id where shop_id is not null			
unique user_id, product_id where product_id is not null			
index user_id			

recently_viewed_items

User browsing history.

Column	Type	Constraint	Purpose
id	uuid	PK	View id
user_id	uuid	FK users nullable	Known user
session_id	varchar(120)	nullable	Anonymous session
shop_id	uuid	FK shops nullable	Viewed shop
product_id	uuid	FK products nullable	Viewed product
viewed_at	timestampz	not null	Time
metadata	jsonb	default {}	Referrer/device

Indexes / Constraints			
index user_id, viewed_at			
index session_id			
index shop_id			
index product_id			

search_queries

Track search behavior for improvements.

Column	Type	Constraint	Purpose
id	uuid	PK	Search id
user_id	uuid	FK users nullable	User
query	text	not null	Search text
filters	jsonb	default {}	Filters
result_count	int	default 0	Results
clicked_shop_id	uuid	FK shops nullable	Clicked shop
clicked_product_id	uuid	FK products nullable	Clicked product

Column	Type	Constraint	Purpose
created_at	timestampz	not null	Searched time
Indexes / Constraints			
index created_at			
GIN index query text			
index clicked_shop_id			

reviews

User reviews for shops/products.

Column	Type	Constraint	Purpose
id	uuid	PK	Review id
user_id	uuid	FK users	Reviewer
shop_id	uuid	FK shops nullable	Shop reviewed
product_id	uuid	FK products nullable	Product reviewed
rating	smallint	1-5	Rating
title	varchar(160)	nullable	Review title
body	text	nullable	Review text
status	varchar(30)	default published	pending/published/hidden/rejected
created_at	timestampz	not null	Created
updated_at	timestampz	not null	Updated
deleted_at	timestampz	nullable	Soft delete
Indexes / Constraints			
unique user_id, shop_id where shop_id is not null and deleted_at is null			
unique user_id, product_id where product_id is not null and deleted_at is null			
index shop_id, status			
index product_id, status			
index rating			

review_replies

Shop/admin replies to reviews.

Column	Type	Constraint	Purpose
id	uuid	PK	Reply id
review_id	uuid	FK reviews	Review
user_id	uuid	FK users	Replier

Column	Type	Constraint	Purpose
body	text	not null	Reply text
status	varchar(30)	default published	published/hidden
created_at	timestampz	not null	Created
updated_at	timestampz	not null	Updated
Indexes / Constraints			
index review_id			

review_reports

Reports against abusive reviews.

Column	Type	Constraint	Purpose
id	uuid	PK	Report id
review_id	uuid	FK reviews	Review
reported_by_id	uuid	FK users	Reporter
reason	varchar(120)	not null	Spam/offensive/fake
details	text	nullable	Details
status	varchar(30)	default open	open/reviewed/dismissed/actioned
created_at	timestampz	not null	Created
Indexes / Constraints			
index review_id			
index status			

Orders Ready Commerce

carts

Shopping cart foundation.

Column	Type	Constraint	Purpose
id	uuid	PK	Cart id
user_id	uuid	FK users nullable	User
session_id	varchar(120)	nullable	Anonymous
status	varchar(30)	default active	active/converted/abandoned
currency_code	char(3)	default INR	Currency
created_at	timestampz	not null	Created
updated_at	timestampz	not null	Updated

Indexes / Constraints
index user_id
index session_id
index status

cart_items

Items inside cart.

Column	Type	Constraint	Purpose
id	uuid	PK	Cart item id
cart_id	uuid	FK carts	Cart
shop_id	uuid	FK shops	Shop
product_id	uuid	FK products	Product
variant_id	uuid	FK variants nullable	Variant
quantity	int	not null default 1	Quantity
unit_price	numeric(12,2)	not null	Price snapshot
metadata	jsonb	default {}	Options
created_at	timestamptz	not null	Created

Indexes / Constraints
unique cart_id, variant_id
index cart_id
index shop_id

orders

Customer order or enquiry order.

Column	Type	Constraint	Purpose
id	uuid	PK	Order id
order_number	varchar(40)	unique	Human order number
user_id	uuid	FK users	Customer
shop_id	uuid	FK shops	Shop
status	varchar(40)	default pending	pending/confirmed/ready/completed/cancelled/refunded
fulfillment_type	varchar(40)	default pickup	pickup/delivery/enquiry
subtotal_amount	numeric(12,2)	not null	Subtotal
tax_amount	numeric(12,2)	default 0	Tax
discount_amount	numeric(12,2)	default 0	Discount

Column	Type	Constraint	Purpose
total_amount	numeric(12,2)	not null	Total
currency_code	char(3)	default INR	Currency
customer_note	text	nullable	Note
metadata	jsonb	default {}	Extra
created_at	timestamptz	not null	Created
updated_at	timestamptz	not null	Updated
Indexes / Constraints			
unique order_number			
index user_id			
index shop_id, status			
index created_at			

order_items

Order item snapshots.

Column	Type	Constraint	Purpose
id	uuid	PK	Order item id
order_id	uuid	FK orders	Order
product_id	uuid	FK products nullable	Product ref
variant_id	uuid	FK variants nullable	Variant ref
product_title	varchar(220)	not null	Snapshot title
variant_title	varchar(180)	nullable	Snapshot variant
sku	varchar(120)	nullable	Snapshot SKU
quantity	int	not null	Quantity
unit_price	numeric(12,2)	not null	Unit price
tax_amount	numeric(12,2)	default 0	Tax
line_total	numeric(12,2)	not null	Line total
Indexes / Constraints			
index order_id			
index product_id			

order_status_history

Order status audit.

Column	Type	Constraint	Purpose
id	uuid	PK	History id
order_id	uuid	FK orders	Order
from_status	varchar(40)	nullable	Old status
to_status	varchar(40)	not null	New status
note	text	nullable	Note
changed_by_id	uuid	FK users nullable	Actor
created_at	timestampz	not null	Time

Indexes / Constraints
index order_id
index created_at

payment_transactions

Payment record foundation.

Column	Type	Constraint	Purpose
id	uuid	PK	Payment id
order_id	uuid	FK orders nullable	Order
shop_invoice_id	uuid	FK shop_commission_invoices nullable	Commission invoice
provider	varchar(60)	not null	stripe/razorpay/cash/manual
provider_payment_id	varchar(180)	nullable	Gateway id
amount	numeric(12,2)	not null	Amount
currency_code	char(3)	default INR	Currency
status	varchar(40)	default pending	pending/succeeded/failed/refunded
paid_at	timestampz	nullable	Paid time
metadata	jsonb	default {}	Gateway payload
created_at	timestampz	not null	Created

Indexes / Constraints
index order_id
index shop_invoice_id
index provider_payment_id
index status

Commission, Billing & Shop Payments

commission_plans

Platform commission plan catalog.

Column	Type	Constraint	Purpose
id	uuid	PK	Plan id
name	varchar(120)	not null	Plan name
code	varchar(80)	unique	standard/premium/custom
commission_type	varchar(30)	not null	percentage/fixed/hybrid
commission_rate	numeric(5,2)	default 0	Percent rate
fixed_amount	numeric(12,2)	default 0	Fixed amount
billing_cycle	varchar(30)	default monthly	monthly/weekly/yearly/manual
grace_period_days	int	default 7	Grace days
features	jsonb	default {}	Plan capabilities
is_active	boolean	default true	Active
created_at	timestamptz	not null	Created
Indexes / Constraints			
unique code			
index is_active			

shop_commission_settings

Active commission configuration for each shop.

Column	Type	Constraint	Purpose
id	uuid	PK	Setting id
shop_id	uuid	FK shops unique	Shop
commission_plan_id	uuid	FK commission_plans	Plan
custom_commission_rate	numeric(5,2)	nullable	Override percent
custom_fixed_amount	numeric(12,2)	nullable	Override fixed
billing_cycle	varchar(30)	nullable	Override cycle
status	varchar(40)	default active	active/paused/overdue/cancelled
auto_pause_on_overdue	boolean	default true	Pause shop if unpaid
next_invoice_date	date	nullable	Next billing date
created_at	timestamptz	not null	Created
updated_at	timestamptz	not null	Updated
Indexes / Constraints			
unique shop_id			

Indexes / Constraints
index status
index next_invoice_date

shop_commission_invoices

Monthly/periodic invoice to shop owner.

Column	Type	Constraint	Purpose
id	uuid	PK	Invoice id
shop_id	uuid	FK shops	Shop
invoice_number	varchar(50)	unique	Human invoice number
billing_period_start	date	not null	Period start
billing_period_end	date	not null	Period end
subtotal_amount	numeric(12,2)	not null	Subtotal
tax_amount	numeric(12,2)	default 0	Tax
discount_amount	numeric(12,2)	default 0	Discount
total_amount	numeric(12,2)	not null	Total
amount_paid	numeric(12,2)	default 0	Paid amount
status	varchar(40)	default draft	draft/issued/paid/overdue/void
due_date	date	not null	Due date
issued_at	timestamptz	nullable	Issued
paid_at	timestamptz	nullable	Paid
metadata	jsonb	default {}	Extra
created_at	timestamptz	not null	Created

Indexes / Constraints
unique invoice_number
index shop_id, status
index due_date
unique shop_id, billing_period_start, billing_period_end

shop_invoice_items

Line items for commission invoice.

Column	Type	Constraint	Purpose
id	uuid	PK	Item id
invoice_id	uuid	FK shop_commission_invoices	Invoice

Column	Type	Constraint	Purpose
item_type	varchar(60)	not null	commission/subscription/penalty/discount/tax
description	text	not null	Line description
quantity	numeric(12,2)	default 1	Quantity
unit_amount	numeric(12,2)	not null	Unit amount
line_total	numeric(12,2)	not null	Line total
metadata	jsonb	default {}	Order refs/details
Indexes / Constraints			
index invoice_id			
index item_type			

shop_payments

Manual/offline payments by shop for commission invoices.

Column	Type	Constraint	Purpose
id	uuid	PK	Shop payment id
shop_id	uuid	FK shops	Shop
invoice_id	uuid	FK shop_commission_invoices nullable	Invoice
amount	numeric(12,2)	not null	Paid amount
payment_method	varchar(60)	not null	upi/bank/cash/card/manual
reference_number	varchar(160)	nullable	Payment reference
proof_media_id	uuid	FK media_assets nullable	Proof screenshot
status	varchar(40)	default pending_review	pending_review/approved/rejected
received_by_id	uuid	FK users nullable	Admin
received_at	timestampz	nullable	Received time
notes	text	nullable	Notes
created_at	timestampz	not null	Created
Indexes / Constraints			
index shop_id			
index invoice_id			
index status			

Support & Communication

support_tickets

Support tickets from users or shop owners.

Column	Type	Constraint	Purpose
id	uuid	PK	Ticket id
ticket_number	varchar(40)	unique	Human ticket no
created_by_id	uuid	FK users	Requester
shop_id	uuid	FK shops nullable	Related shop
assigned_to_id	uuid	FK users nullable	Support/admin assignee
category	varchar(60)	not null	billing/shop/product/technical/general
priority	varchar(30)	default medium	low/medium/high/urgent
status	varchar(40)	default open	open/in_progress/waiting/resolved/closed
subject	varchar(220)	not null	Subject
description	text	not null	Issue
resolved_at	timestamptz	nullable	Resolved time
created_at	timestamptz	not null	Created
updated_at	timestamptz	not null	Updated

Indexes / Constraints

unique ticket_number

index created_by_id

index shop_id

index status, priority

index assigned_to_id

support_messages

Conversation messages in ticket.

Column	Type	Constraint	Purpose
id	uuid	PK	Message id
ticket_id	uuid	FK support_tickets	Ticket
sender_id	uuid	FK users	Sender
body	text	not null	Message
is_internal_note	boolean	default false	Admin-only note
created_at	timestamptz	not null	Created

Indexes / Constraints

index ticket_id, created_at

Indexes / Constraints

index sender_id

support_attachments

Attachments for support messages.

Column	Type	Constraint	Purpose
id	uuid	PK	Attachment id
message_id	uuid	FK support_messages	Message
media_id	uuid	FK media_assets	File
created_at	timestampz	not null	Created

Indexes / Constraints

index message_id

notification_templates

Reusable notification templates.

Column	Type	Constraint	Purpose
id	uuid	PK	Template id
code	varchar(100)	unique	shop_approved/payment_due
channel	varchar(30)	not null	email/sms/push/in_app
subject	varchar(220)	nullable	Email subject
body	text	not null	Template body
variables	jsonb	default []	Allowed variables
is_active	boolean	default true	Enabled
created_at	timestampz	not null	Created

Indexes / Constraints

unique code, channel

index is_active

notifications

In-app/push/email notification records.

Column	Type	Constraint	Purpose
id	uuid	PK	Notification id
user_id	uuid	FK users	Recipient

Column	Type	Constraint	Purpose
template_id	uuid	FK templates nullable	Template
channel	varchar(30)	not null	in_app/email/sms/push
title	varchar(220)	nullable	Title
body	text	not null	Body
data	jsonb	default {}	Deep link/payload
status	varchar(30)	default pending	pending/sent/read/failed
read_at	timestamptz	nullable	Read time
sent_at	timestamptz	nullable	Sent time
created_at	timestamptz	not null	Created
Indexes / Constraints			
index user_id, status			
index created_at			
index channel			

System, Audit & Operations

audit_logs

Security/business audit trail.

Column	Type	Constraint	Purpose
id	uuid	PK	Audit id
actor_user_id	uuid	FK users nullable	Actor
action	varchar(120)	not null	shops.approved/product.updated
entity_type	varchar(80)	not null	shops/products/users
entity_id	uuid	nullable	Affected id
old_values	jsonb	nullable	Before
new_values	jsonb	nullable	After
ip_address	inet	nullable	IP
user_agent	text	nullable	UA
created_at	timestamptz	not null	Created
Indexes / Constraints			
index actor_user_id			
index entity_type, entity_id			
index action			
index created_at			

app_settings

Key-value app configuration.

Column	Type	Constraint	Purpose
id	uuid	PK	Setting id
key	varchar(120)	unique	Setting key
value	jsonb	not null	Setting value
description	text	nullable	Meaning
is_public	boolean	default false	Safe for frontend
updated_by_id	uuid	FK users nullable	Updater
updated_at	timestampz	not null	Updated
Indexes / Constraints			
unique key			
index is_public			

feature_flags

Runtime feature toggles.

Column	Type	Constraint	Purpose
id	uuid	PK	Flag id
key	varchar(120)	unique	Flag key
description	text	nullable	Purpose
is_enabled	boolean	default false	Enabled globally
rules	jsonb	default {}	Targeting rules
created_at	timestampz	not null	Created
updated_at	timestampz	not null	Updated
Indexes / Constraints			
unique key			
index is_enabled			

import_jobs

Bulk imports for products/shops/categories.

Column	Type	Constraint	Purpose
id	uuid	PK	Job id

Column	Type	Constraint	Purpose
created_by_id	uuid	FK users	Creator
job_type	varchar(60)	not null	product_import/shop_import
source_media_id	uuid	FK media_assets nullable	CSV/XLSX file
status	varchar(40)	default queued	queued/running/completed/failed
total_rows	int	default 0	Rows
processed_rows	int	default 0	Processed
failed_rows	int	default 0	Failed
error_summary	text	nullable	Error
result	jsonb	default {}	Details
created_at	timestampz	not null	Created
completed_at	timestampz	nullable	Completed
Indexes / Constraints			
index created_by_id			
index status			
index job_type			

webhook_events

External payment/app webhook idempotency.

Column	Type	Constraint	Purpose
id	uuid	PK	Webhook id
provider	varchar(60)	not null	stripe/razorpay/etc
event_id	varchar(180)	not null	Provider event id
event_type	varchar(120)	not null	Event type
payload	jsonb	not null	Raw payload
status	varchar(40)	default received	received/processed/failed/ignored
processed_at	timestampz	nullable	Processed
created_at	timestampz	not null	Received
Indexes / Constraints			
unique provider, event_id			
index event_type			
index status			

4. Critical Relationships

Parent	Child	Cardinality	Delete rule / notes
users	shops.owner_user_id	1 -> many	Restrict delete; soft delete users.
users	shop_team_members.user_id	1 -> many	A user can staff multiple shops.
shops	shop_locations	1 -> many	Cascade soft business behavior; DB restrict or cascade depending ORM.
shops	shop_business_hours	1 -> 7	Exactly one row per weekday.
shops	shop_documents	1 -> many	Verification files.
categories	categories.parent_id	self tree	Parent category hierarchy.
categories	shop_categories	many -> many	Shop can have multiple categories.
categories	product_category_links	many -> many	Product can have multiple categories.
shops	products	1 -> many	Shop owns products.
products	product_variants	1 -> many	Variants optional; simple products can still have default variant.
product_variants	product_inventory	1 -> many	Per branch/location inventory.
shops	shop_commission_settings	1 -> 1	One active billing setting per shop.
shops	shop_commission_invoices	1 -> many	Monthly/periodic invoices.
support_tickets	support_messages	1 -> many	Ticket conversation.
orders	order_items	1 -> many	Snapshot values for order history.

5. Enum Values

Enum	Values
shop.status	draft, pending_review, active, paused, suspended, rejected, closed
shop.verification_status	unverified, pending, verified, rejected
shop.commission_status	not_configured, active, overdue, paused
product.status	draft, active, out_of_stock, hidden, archived, rejected
user.status	active, pending, suspended, deleted
support_ticket.status	open, in_progress, waiting, resolved, closed
invoice.status	draft, issued, paid, overdue, void
payment.status	pending, succeeded, failed, refunded
review.status	pending, published, hidden, rejected
order.status	pending, confirmed, ready, completed, cancelled, refunded

6. Indexing Strategy

- **Search:** products should have a PostgreSQL full-text search vector/index across title, description, tags, search_keywords, shop name, and category names. Start with generated tsvector or query-time vector, then move to Meilisearch/Typesense later if needed.
- **Location:** shop_locations should use PostGIS geography Point if available. If not, store latitude/longitude and add btree indexes; upgrade to PostGIS for radius search.

- **Soft delete uniqueness:** use partial unique indexes such as unique slug where deleted_at is null.
- **Admin lists:** index status columns on shops, products, support_tickets, invoices, reviews.
- **Foreign keys:** every *_id used for joins must have an index unless it is already part of a unique constraint.
- **JSONB:** do not over-index metadata initially; only add GIN indexes when a JSONB field becomes a real filter.

7. MVP Build Order for Backend

Phase	Build tables/modules	Why first
1. Foundation	users, roles, permissions, media_assets, audit_logs, app_settings	Auth, access, file upload, admin traceability.
2. Location	countries, states, cities, areas, user_addresses	Needed before shop creation and search.
3. Shop Core	shops, shop_locations, shop_business_hours, shop_documents, shop_team_members, shop_status_history	Admin and shop owner workflow.
4. Catalog	categories, shop_categories, brands, products, variants, inventory, product_media	Main Localo product value.
5. Discovery	favorites, recently_viewed_items, search_queries, reviews	User app value and feedback loop.
6. Monetization	commission_plans, settings, invoices, invoice_items, shop_payments	Shop commission flow and pause-on-unpaid logic.
7. Support	support_tickets, messages, attachments, notifications	Operational support.
8. Commerce-ready	carts, orders, order_items, payments	Add when Localo moves beyond discovery/enquiry.

8. Prisma/PostgreSQL Implementation Notes

- Use Prisma enum types or string enums. If using PostgreSQL enums, plan migrations carefully because enum edits are annoying later.
- Prefer Decimal in Prisma for money fields, never Float.
- Use @db.Timestamptz(6) for timestamps if available.
- Use Unsupported("geography(Point,4326)") for PostGIS geography columns if Prisma cannot model them directly; manage index in SQL migration.
- Keep deletedAt nullable on business tables and filter deletedAt: null in repository/service layer.
- Seed roles, permissions, default categories, commission plans, app settings, and demo areas first.

9. Open Decisions Before Coding

Decision	Recommended default	Can change later?
Will Localo take orders or only discovery first?	Discovery + enquiry first, keep orders-ready tables for later.	Yes
Commission model	Monthly invoice per shop + optional percentage later.	Yes
Admin creates shops vs shop self-registration	Admin creates shops first; shop owner receives/invited.	Yes
Location search	PostGIS if possible; otherwise lat/lng columns now.	Yes

Decision	Recommended default	Can change later?
Category as array or relation	Use relations for real filtering; keep product.tags text[] for quick search labels.	Yes
Inventory required?	Optional tracking per product/variant.	Yes
Payments now?	Manual payment records first; gateway later.	Yes

10. Codex Implementation Prompt Plan

When Codex is available, do not ask it to build everything at once. Use focused prompts like these:

1. Create Prisma schema for Identity, RBAC, Media, and Audit only based on the PDF.
2. Add Location and Shop Core models with indexes and relations.
3. Add Category and Catalog models with product variants, media, inventory, and attributes.
4. Add Commission/Billing models and services for invoice generation and overdue shop pause.
5. Add Support, Notifications, Reviews, and Discovery tables.
6. Add seed files for roles, permissions, categories, commission plans, and app settings.

End of document. This is a complete draft schema for review before backend implementation. After review, convert final decisions into Prisma models, SQL indexes, migrations, seeders, and API contracts.